```
import kagglehub
import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
path = kagglehub.dataset_download("dhairyajeetsingh/ecommerce-customer-behavior-dataset")

print("Path to dataset files:", path)
print(os.listdir(path))
```

```
Downloading to /root/.cache/kagglehub/datasets/dhairyajeetsingh/ecommerce-customer-behavior-dataset/1.archive...
100%|██████████| 1.96M/1.96M [00:01<00:00, 1.41MB/s]Extracting files...
Path to dataset files: /root/.cache/kagglehub/datasets/dhairyajeetsingh/ecommerce-customer-behavior-dataset/versions/1
['ecommerce_customer_churn_dataset.csv']
```

```
file_path = os.path.join(path, "ecommerce_customer_churn_dataset.csv")
df = pd.read_csv(file_path)

print(df.head())
```

```
    Age  Gender Country        City  Membership_Years  Login_Frequency  \
0  43.0    Male  France   Marseille               2.9             14.0
1  36.0    Male      UK  Manchester               1.6             15.0
2  45.0  Female  Canada   Vancouver               2.9             10.0
3  56.0  Female     USA    New York               2.6             10.0
4  35.0    Male   India       Delhi               3.1             29.0

   Session_Duration_Avg  Pages_Per_Session  Cart_Abandonment_Rate  \
0                  27.4                6.0                   50.6
1                  42.7               10.3                   37.7
2                  24.8                1.6                   70.9
3                  38.4               14.8                   41.7
4                  51.4                NaN                   19.1

   Wishlist_Items  ...  Email_Open_Rate  Customer_Service_Calls  \
0             3.0  ...             17.9                     9.0
1             1.0  ...             42.8                     7.0
2             1.0  ...              0.0                     4.0
3             9.0  ...             41.4                     2.0
4             9.0  ...             37.9                     1.0

   Product_Reviews_Written  Social_Media_Engagement_Score  Mobile_App_Usage  \
0                      4.0                           16.3              20.8
1                      3.0                            NaN              23.3
2                      1.0                            NaN               8.8
3                      5.0                           85.9              31.0
4                     11.0                           83.0              50.4

   Payment_Method_Diversity  Lifetime_Value  Credit_Balance  Churned  \
0                       1.0          953.33          2278.0        0
1                       3.0         1067.47          3028.0        0
2                       NaN         1289.75          2317.0        0
3                       3.0         2340.92          2674.0        0
4                       4.0         3041.29          5354.0        0

   Signup_Quarter
0              Q1
1              Q4
2              Q4
3              Q1
4              Q4

[5 rows x 25 columns]
```

Data Cleaning

```
df.isnull().sum()
null_percent = df.isnull().mean() * 100
print(null_percent.sort_values(ascending=False))
```

```
Social_Media_Engagement_Score    12.000
Credit_Balance                   11.000
Mobile_App_Usage                 10.000
Returns_Rate                      8.982
Wishlist_Items                    8.000
Product_Reviews_Written           7.000
Discount_Usage_Rate               7.000
Session_Duration_Avg              6.798
Pages_Per_Session                 6.000
Days_Since_Last_Purchase          6.000
Email_Open_Rate                   5.056
```

```
Payment_Method_Diversity          5.000
Age                               4.990
Customer_Service_Calls            0.336
Gender                            0.000
Country                           0.000
Membership_Years                  0.000
Cart_Abandonment_Rate             0.000
Login_Frequency                   0.000
City                              0.000
Average_Order_Value               0.000
Total_Purchases                   0.000
Lifetime_Value                    0.000
Churned                           0.000
Signup_Quarter                    0.000
dtype: float64
```

```
#1-5%
columns_to_impute = ['Customer_Service_Calls', 'Age', 'Payment_Method_Diversity', 'Email_Open_Rate']
for col in columns_to_impute:
    df[col] = df[col].fillna(df[col].median())

#5-10%
df['Activity_Tier'] = pd.qcut(df['Login_Frequency'], q=5, labels=[1, 2, 3, 4, 5])
column_to_impute1 = ['Session_Duration_Avg', 'Product_Reviews_Written', 'Wishlist_Items', 'Pages_Per_Session']
for col in column_to_impute1:
    df[col] = df.groupby('Activity_Tier', observed=False)[col].transform(lambda x: x.fillna(x.median()))

df['Membership_Tier'] = pd.qcut(df['Membership_Years'], q=3, labels=['New_User', 'Regular_User', 'Loyal_User'])
df['Discount_Usage_Rate'] = df.groupby('Membership_Tier', observed=False)['Discount_Usage_Rate'].transform(lambda x: x.fil
df['Returns_Rate'] = df.groupby('Customer_Service_Calls')['Returns_Rate'].transform(lambda x: x.fillna(x.median()))
df['Days_Since_Last_Purchase'] = df.groupby('Churned')['Days_Since_Last_Purchase'].transform(lambda x: x.fillna(x.median()

#10-12%
df['SM_missing'] = df['Social_Media_Engagement_Score'].isnull().astype(int)
df['App_missing'] = df['Mobile_App_Usage'].isnull().astype(int)
df['Credit_missing'] = df['Credit_Balance'].isnull().astype(int)

high_missing_cols = ['Social_Media_Engagement_Score', 'Mobile_App_Usage']
for col in high_missing_cols:
    df[col] = df.groupby('Activity_Tier', observed=False)[col].transform(lambda x: x.fillna(x.median()))

df['Credit_Balance'] = df['Credit_Balance'].fillna(0)

df.drop(columns=['Membership_Tier'], inplace=True)
df.drop(columns=['Activity_Tier'], inplace=True)
```

```
df.describe()
```

	Age	Membership_Years	Login_Frequency	Session_Duration_Avg	Pages_Per_Session	Cart_Abandonment_Rate	Wish
count	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50000.000000	50
mean	37.812800	2.984009	11.624660	27.353578	8.735973	57.079973	
std	11.535688	2.059105	7.810657	10.647001	3.711174	16.282723	
min	5.000000	0.100000	0.000000	1.000000	1.000000	0.000000	
25%	30.000000	1.400000	6.000000	19.400000	6.000000	46.400000	
50%	38.000000	2.500000	11.000000	26.400000	8.400000	58.100000	
75%	45.000000	4.000000	17.000000	34.100000	11.100000	68.700000	
max	200.000000	10.000000	46.000000	75.600000	24.100000	143.743350	

8 rows × 24 columns

```
df = df[(df['Age'] >= 12) & (df['Age'] <= 90)]
df = df[df['Total_Purchases'] >= 0]
df = df[df['Cart_Abandonment_Rate'] <= 100]

# Tangani Outlier AOV menggunakan metode memangkas nilai ekstrem (1% teratas)
q_high = df['Average_Order_Value'].quantile(0.99)
df = df[df['Average_Order_Value'] < q_high]
```

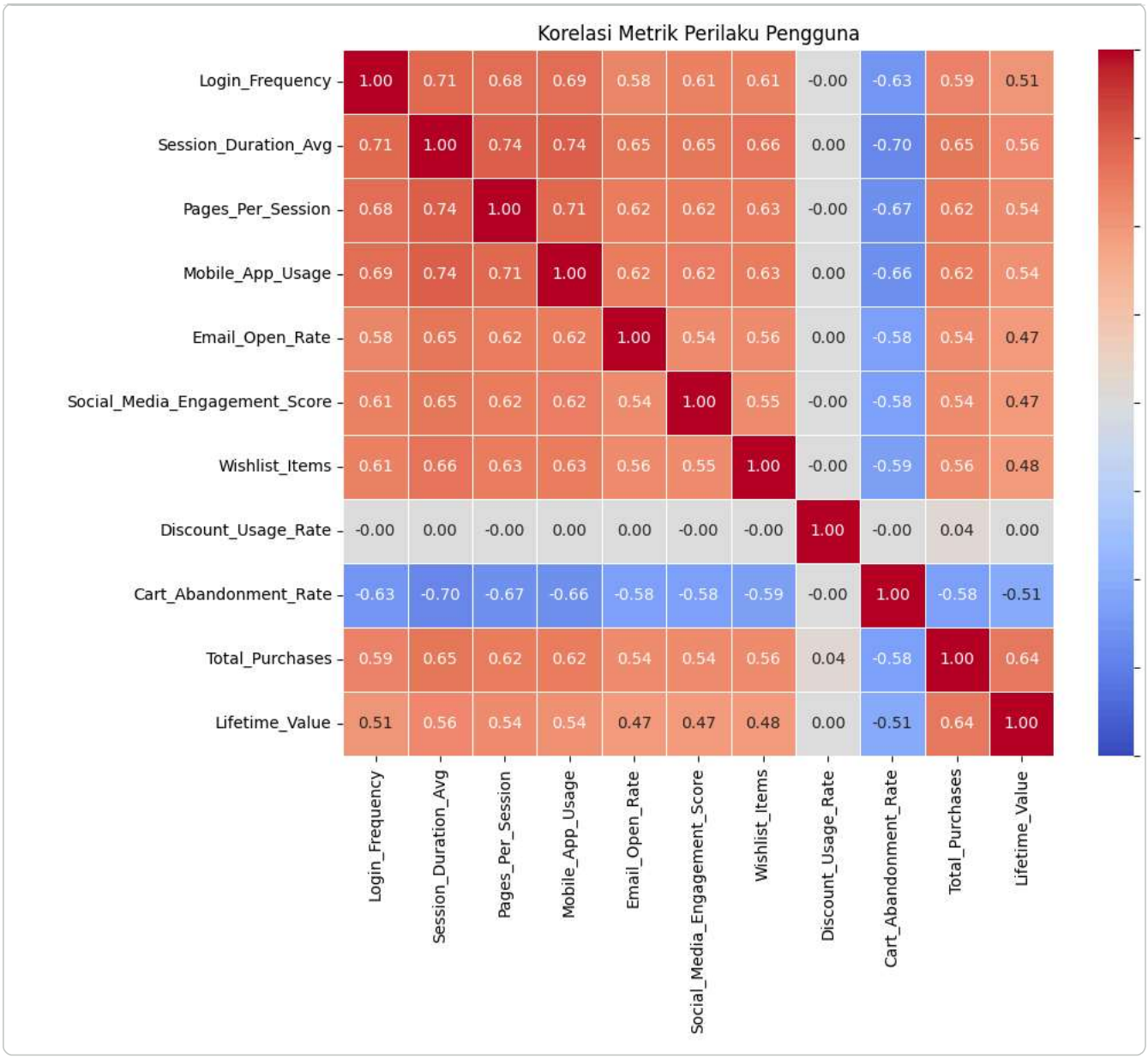
```
df.describe()
```

	Age	Membership_Years	Login_Frequency	Session_Duration_Avg	Pages_Per_Session	Cart_Abandonment_Rate	Wish
count	49381.000000	49381.000000	49381.000000	49381.000000	49381.000000	49381.000000	49
mean	37.782609	2.982930	11.630202	27.361710	8.740392	57.033229	
std	11.165671	2.057075	7.814457	10.651151	3.710558	16.213993	
min	18.000000	0.100000	0.000000	1.000000	1.000000	0.000000	
25%	30.000000	1.400000	6.000000	19.400000	6.000000	46.400000	
50%	38.000000	2.500000	11.000000	26.400000	8.400000	58.100000	
75%	45.000000	4.000000	17.000000	34.200000	11.100000	68.700000	
max	75.000000	10.000000	46.000000	75.600000	24.100000	100.000000	

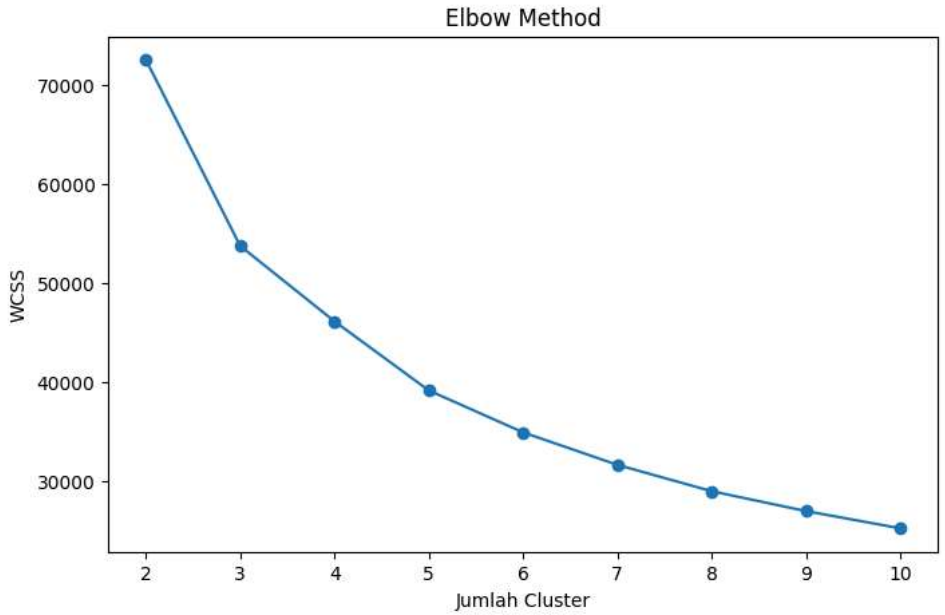
8 rows × 24 columns

User Behavior

```
behavior_metrics = ['Login_Frequency',
    'Session_Duration_Avg',
    'Pages_Per_Session',
    'Mobile_App_Usage',
    'Email_Open_Rate',
    'Social_Media_Engagement_Score',
    'Wishlist_Items',
    'Discount_Usage_Rate',
    'Cart_Abandonment_Rate',
    'Total_Purchases',
    'Lifetime_Value']
correlation_matrix = df[behavior_metrics].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt=".2f", linewidths=0.5, vmin=-1, vmax=1)
plt.title('Korelasi Metrik Perilaku Pengguna')
plt.show()
```




```
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans

behavior_features = ['Login_Frequency', 'Session_Duration_Avg', 'Total_Purchases']
X = df[behavior_features]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

wcss = []
for i in range(2,11):
    kmeans = KMeans(n_clusters=i, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)
plt.figure(figsize=(8,5))
plt.plot(range(2,11), wcss, marker='o')
plt.xlabel("Jumlah Cluster")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()

kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
df['User_Segment'] = kmeans.fit_predict(X_scaled)

segment_mapping = {0: 'Segment 1', 1: 'Segment 2', 2: 'Segment 3'}
df['User_Segment'] = df['User_Segment'].map(segment_mapping)

segment_profile = df.groupby('User_Segment')[['Login_Frequency', 'Session_Duration_Avg', 'Total_Purchases', 'Lifetime_Value'
print(segment_profile.round(2))
```




```
              Login_Frequency  Session_Duration_Avg  Total_Purchases  \
User_Segment
Segment 1                5.08                 18.36             8.06
Segment 2               13.37                 29.56            14.02
Segment 3               22.47                 42.66            22.53

              Lifetime_Value  Days_Since_Last_Purchase  Cart_Abandonment_Rate  \
User_Segment
Segment 1             889.75                     28.95                  68.00
Segment 2            1546.83                     29.55                  54.60
Segment 3            2341.77                     29.42                  37.85

              Churned
User_Segment
Segment 1        0.40
Segment 2        0.21
Segment 3        0.20
```

```
segment_mapping = {'Segment 1':'Low Activity User', 'Segment 2':'Moderately Active Users', 'Segment 3':'High Activity Users
df['User_Segment'] = df['User_Segment'].replace(segment_mapping)
```

```
df['User_Segment'].value_counts()
```

User_Segment	count
Low Activity User	20363
Moderately Active Users	19900
High Activity Users	9118

dtype: int64

```
profile = df.groupby('User_Segment')[
    behavior_features
].mean()

profile.plot(
    kind='bar',
    figsize=(10,6)
)

plt.title("Rata-rata Perilaku Setiap Segment")
plt.ylabel("Average")
plt.xticks(rotation=0)
plt.show()
```